Exercise Session 1: Organization and Project Assignment

Automatic Image Analysis

Exercise Session 1

Organization, project assignment, and semester workflow.

- Automatic Image Analysis
- Introductory exercise meeting and topic selection

Computer Vision and Remote Sensing Group, TU Berlin
Summer Semester 2026

Contact and consultation

Contact

Name: Annamalai Lakshmanan

Email: `annamalai.lakshmanan@tu-berlin.de`

Office: Room 6.048, MAR Building

Exercise Scheduling

Happens every two weeks, next one on May 4.

Consultation hours

5pm - 6 pm Tuesdays starting next week

Important course expectations

Technical readiness

- Basic coding skills are necessary for this course.
- You should be able to run Python notebooks, inspect data, and debug simple errors.
- Use Google Colab, Kaggle, or any available compute environment for experiments.

Rules that matter

- The exercise is compulsory for passing the exam.
- Plagiarism is treated strictly.
- Deadlines are strict and must be planned for early.
- Every group member must understand the submitted work.

Exercise 1 roadmap

Focus of today

Exercise Session 1 is about structure, expectations, and choosing a project with a realistic plan for the semester.

- Understand how the exercise sessions and milestones are organized.
- See the project directions and what kind of work each one involves.
- Clarify working rules, documentation habits, and responsible tool use.
- Form groups, choose a topic, and leave with a concrete next step.

Agenda for today

Part 1

- Organizational overview
- Rules and expectations
- Semester workflow

Part 2

- Project directions
- Group formation and topic choice
- First practical next steps

By the end of the session, start forming groups and deciding on topics.

How the exercise is organized

Project-based format

- You work on a small project during the term.
- The goal is not only implementation, but also interpretation.
- Visual results, comparisons, and clear reasoning all matter.

What success looks like

- A method that is implemented and tested.
- Results shown on meaningful examples.
- Honest discussion of failures and limitations.
- A clear link between choices and observed effects.

Submission mode and group work

Working mode

- Projects are prepared in small groups.
- Groups typically consist of 3 to 4 students.
- The introductory exercise is the starting point for topic selection.
- Every group member should understand the full pipeline, not only one isolated part.

Recommendation

- Build your environment early.
- Share code and visual outputs within the group.
- Keep responsibilities transparent from the beginning.
- Use a shared folder or repository from week one.

Background and tooling

What is expected

- Basic programming is sufficient.
- The course emphasizes algorithms, not advanced software engineering.
- You should be able to read results and explain behavior.

Recommended setup

- Python with NumPy, Matplotlib, and OpenCV.
- A simple, reproducible folder structure.
- Early testing on a few images before scaling up.

Working rules and expectations

Rules for the semester

- Respect hard deadlines.
- Keep work reproducible and organized.
- Communicate blockers early, not at the last minute.
- Do not hide negative results; discuss them.

Practical habits

- Save figures with clear filenames.
- Record parameter settings while you work.
- Compare methods on the same examples.
- Keep short notes on what changed and why.

Using LLMs responsibly

Working rule

Use LLMs as assistants for brainstorming, debugging, and explanation support, but not as a substitute for your own understanding.

- Do not submit code or text that you cannot explain clearly.
- Verify generated code, equations, and claims on your own examples.
- Use the tools to speed up iteration, not to skip the scientific reasoning.
- If you use substantial assistance, document it transparently in your notes.

Topic assignment overview

Theme

Each topic compares a classical image-analysis pipeline with a frozen or lightly trained modern representation on the same task.

- Topic A: Shape and Contour - Leaf Species Recognition
- Topic B: Bayesian Decision and Uncertainty - Skin Lesion Triage
- Topic C: Scale-Space and Invariant Detection - Matching Across Viewpoints
- Topic D: Self-Supervised Representation Learning - STL-10 Linear Evaluation
- Topic E: Video Understanding - Temporal Action Recognition on UCF-101

Topic A – Shape and Contour: Leaf Species Recognition

At a glance

Dataset: Flavia leaf images, 32 species.

Task: Classify plant species from segmented leaf silhouettes.

What to compare

- Classical: Fourier descriptors and Hu moments with simple classifiers.
- Modern: frozen MobileNetV2 features with a linear SVM.
- Report: confusion matrix, per-class accuracy, and when shape features fail.

Topic B – Bayesian Decision and Uncertainty: Skin Lesion Triage

At a glance

Dataset: HAM10000 dermoscopy images, class-balanced subset.

Task: Melanoma vs. benign triage with uncertainty.

What to compare

- Classical: HSV colour histograms, Gaussian mixtures, and Bayesian decision rules.
- Modern: frozen EfficientNet-B0 with a trained classification head.
- Report: ROC curves, calibration, and uncertainty examples.

Topic C – Scale-Space and Invariant Detection: Matching Across Viewpoints

At a glance

Dataset: VGG Affine Covariant Regions Benchmark.

Task: Match local features under viewpoint changes.

What to compare

- Classical: DoG keypoints with SIFT descriptors.
- Modern: SuperPoint descriptors on the same image pairs.
- Report: repeatability, matching score, and qualitative failure cases.

Topic D – Self-Supervised Representation Learning: STL-10 Linear Evaluation

At a glance

Dataset: STL-10 labelled and unlabelled images.

Task: Test whether pretrained representations reduce the need for labels.

What to compare

- Classical: HOG features with a linear SVM.
- Modern: frozen DINO ViT-S/16 features with the same linear SVM.
- Report: learning curves for different labelled-data fractions.

Topic E – Video Understanding: Temporal Action Recognition on UCF-101

At a glance

Dataset: UCF-101, 10-class subset.

Task: Classify short human-action clips.

What to compare

- Classical: dense optical flow features with a multi-class SVM.
- Modern: frozen ResNet-50 frame features with temporal average pooling.
- Report: accuracy, confusion matrix, and where temporal order matters.

How to choose a project

A good choice is a practical one

Choose the topic that fits your curiosity, your group strengths, and the type of experiments you can realistically carry out over the semester.

- Choose a project whose data you find interesting enough to inspect repeatedly.
- Prefer a topic where you can define simple first experiments immediately.
- Avoid choosing only by difficulty label; choose by what you can analyze well.

Semester workflow

Session 1 → Presentation 1 → Presentation 2 → Final Report

- Two presentations structure the semester and force iterative progress.
- One final report consolidates methods, experiments, and interpretation.

Milestones

Session 1 starts the work, then the two presentations create feedback loops before the final report.

What you should document throughout the term

Keep track of

- Data subsets and example images
- Parameter settings and versions of your code
- Figures that show both successes and failures

Why this matters

- It makes presentations easier to prepare.
- It helps you compare methods fairly.
- It turns experiments into a reportable story.

Deadlines and practical advice

Hard deadline

Late submission means missing the evaluation window, so planning backward from milestones is essential.

Practical advice

- Start data inspection immediately.
- Keep visual results from week to week.
- Record parameters and failure cases.
- Connect every method choice to a measurable effect.

Today: what you should do

Exercise session task

Form groups, choose a topic, and register the assignment on or before **May 3**.

- Start from the data and the main question.
- Pick a topic that matches your interest and available time.
- Decide who will set up the first code skeleton and who will collect initial examples.
- Aim for a method you can test visually from the beginning.
- Leave today with a concrete first experiment and a short division of tasks.